

BRAC UNIVERSITY
Department of Computer Science and Engineering

Examination : Semester Final
 Duration: **2 Hours**

Semester: **Summer 2025**
 Full Marks: **70**

CSE421 / EEE465 : Computer Networks

Answer **Sections A, B and C** as per instructions given. (**Pages: 3**)

Figures in the right margin indicate marks.

Name:

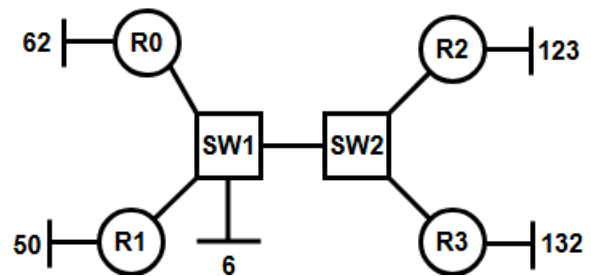
ID:

Section:

SECTION A [All questions of this section are MANDATORY] - 40 MARKS

Q1
[CO3] The number of hosts given in the topology only includes PCs, printers and servers.

Sheldon and Leonard are both on **R1-LAN**. Their IP addresses are **192.0.50.69** and **192.127.255.253** (2nd last usable IP address of the network), respectively. Penny, who lives in the same building, is also on this network and uses the first usable IP address.



- Calculate the subnet mask of the R1-LAN based on the above scenario.
- Penny decides to send a packet to all devices on **R1-LAN**. Which IP address shall she use as the destination?
- Sheldon now plans to expand his office after winning the Nobel Peace Prize and hence designs the topology given above. Apply **VLSM** using the network address **69.0.0.0/15** to efficiently create the subnet addresses for the expanded topology.
- For the switched network, how many addresses are we wasting?

Q2

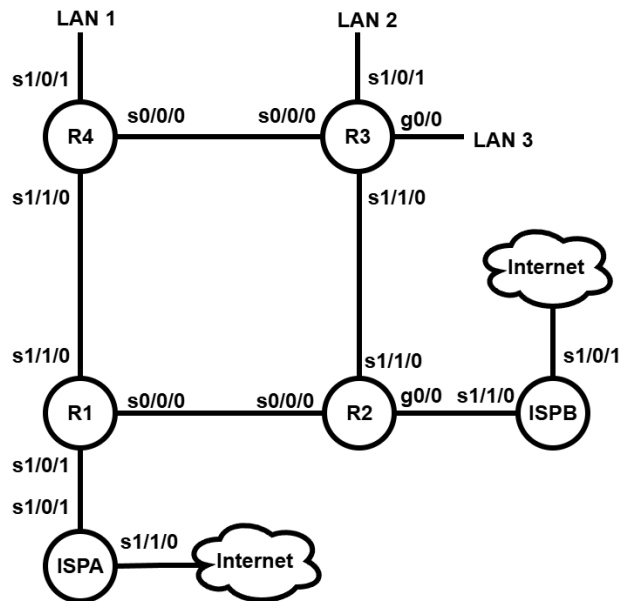
#	Total Length	MF	DF	Offset
...	...	...	...	...
10	1544	1	1	1692
...	...	...	...	...
15	69	0	1	2632

From the above table,

- [CO3]** Identify the header size of the fragment
- [CO3]** Identify the packet size of the original datagram
- [CO2]** Which two fields are required to orderly reassemble the fragmented packets?

Q3

Device	Interface	IP	Network
R1	s0/0/0	.101	172.20.10.100/30
	s1/1/0	.225	172.20.11.224/30
	s1/0/1	.1	172.20.9.0/30
R2	s1/1/0	.193	172.20.12.192/30
	s0/0/0	.102	172.20.10.100/30
	g0/0	.1	178.51.100.0/30
R3	s1/0/1	.1	172.20.96.0/24
	g0/0	.1	172.20.80.0/26
	s1/1/0	.194	172.20.12.192/30
	s0/0/0	.97	172.20.10.96/30
R4	s1/0/1	.1	172.20.64.0/25
	s0/0/0	.98	172.20.10.96/30
	s1/1/0	.226	172.20.11.224/30
ISPA	s1/0/1	.2	172.20.9.0/30
	s1/1/0	.1	210.1.1.0/24
ISPB	s1/1/0	.2	178.51.100.0/30
	s1/0/1	.1	215.1.1.0/24



- [CO3] I. On **R2**, add a **recursive** static route to **LAN1** via **R1**.
- [CO3] II. Configure a **back-up route (directly attached)** of the above static route (AD 6).
- [CO3] III. On **R4**, add a **recursive default** static route to go through the **ISPB** link.
- [CO2] IV. If the above route wasn't configured on **R4**, what problem would have arisen?
- [CO2] V. Two static routes are configured on **R3**: S 172.20.64.0/25 [30/0] via 172.20.10.98 and S 172.20.64.0/25 [20/0] via 172.20.12.193. Which entry should be in the routing table of **R3**, and why?

END OF SECTION A

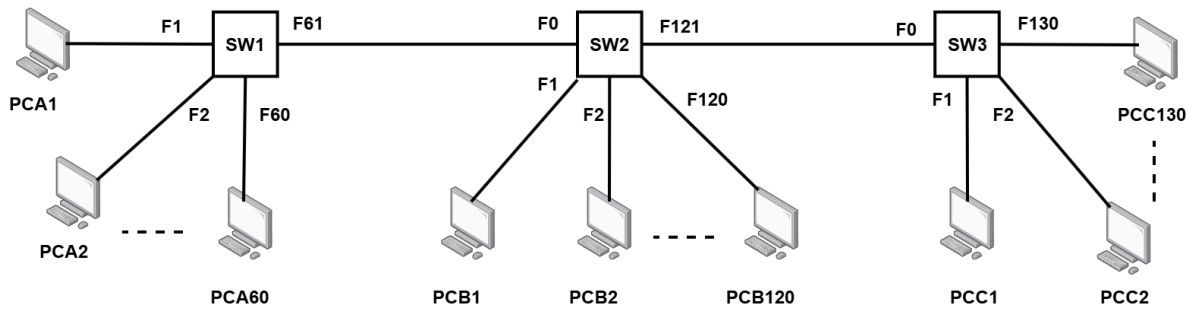
[CO3] **SECTION B** [Answer ANY TWO out of THREE in this section] - 12 MARKS

Q4 We're running RIP on the topology mentioned in **Q3**. All the routers were simultaneously in their cold start at $t = 0$ seconds.

- At the cold start, which networks' information will be present in the **R4** router's routing table?
- After the network has converged, the **R3–R2** link fails. **R3** updates its distance to **ISPB**. **Identify** what information **R3** will share to convey the change, and to which interfaces does **R3** advertise this change?

- Q5**
- Apply** your knowledge of IPv6 address formatting to transform and **simplify** the following addresses into their shortest possible form according to IPv6 rules.
 - 2001:0db8:12af:0000:0000:0000:0a20:0004
 - 0000:0000:0000:0000:0000:0000:0000:0001.
 - For (B), identify the type of address.
 - There is no broadcast address in IPv6. **Explain** with an example how a device will send a packet to all the devices of its network using IPv6.

Q6

3
+3

Consider a network, shown in the above figure, in which PCA1 connected to SW1 wants to send an ARP request frame for PCC2, connected to SW3. Initially, SW3 has the information of PCC2 connected to it.

- I. How many PCs will receive the ARP request frame? Why?
- II. Draw the switch tables of all 3 switches once PCA1 receives the reply.

END OF SECTION B

[CO2] SECTION C [Answer **ANY THREE out of FIVE** in this section] - 18 MARKS

Q7

- I. In IPv4 fragmentation, why is the fragment offset measured in multiples of 8 bytes instead of single bytes?
- II. Why is IPv4 unreliable?
- III. In which scenarios will traceroute help more than PING?

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2
+
2

Q8

A company has a centralized DHCP server located in its headquarters. A new branch office connects its PCs to the network, but users complain that their machines cannot get IP addresses automatically. The branch router is not running a DHCP server.

- I. Explain why the PCs cannot obtain IP addresses in this setup.
- II. Identify the role of the branch router in solving this issue.

3
+
3

Q9

You have set up a gaming server and given it the name www.wowgame.com. The server uses a public IP address from your ISP, but this IP is shared with other customers.

- I. Why can't your friends from abroad connect to your gaming server?
- II. What networking solution would allow your server to be reached directly from the Internet?

3
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3

Q10

- I. Explain how IPv6 packets can still travel across an IPv4-only network. Why is this method useful during the transition period when both IPv4 and IPv6 are in use?
- II. What possible problems can happen when packets are repeatedly wrapped inside other packets and then unwrapped again?

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2

Q11

A PC (192.168.10.50) prints to its network printer (192.168.10.200, MAC: 0C-43-00-12-BC-DD). The printer has been connected to another interface of the switch.

3
+

Initially, the PC cannot print, but after approximately 4 minutes, printing works automatically.	3
I. Why couldn't the PC print immediately after the printer was connected to another port of the switch?	
II. What could have been done to enable printing immediately after the printer was moved to another interface?	

END OF SECTION C

===== **THE END** =====

Why did the IP packet break up with the router?

Too many hops in the relationship.